Using Alloy to Model Agentic Programming

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This paper presents an Alloy model [1] for an AI agent program where the agent can read and write to files, and create pull requests (PRs). The goal of this paper is to show the benefits of formal methods (FM) applied to agentic programs. This paper is divided the following way:

A problem one can highlight is the separation between the purpose of the AI Agent and the tools it has available. Tools are concrete abstractions of what the agent is allowed to do, while a purpose is a natural-language description of the goal. Here lies the problem: Natural-language is ambiguous. We redefine purpose to the less ambiguous noun operation, more specifically an agent has a set of operations. We define an operation as a specific task that lies within the allowable actions a tool has. We can summarize in a more general way: Agentic programs rely on natural-language leading to ambiguous requirements.

There are news, articles, blogs, etc. about the problems created by under specified purposes.

FM can be a potential solution by defining a more strict language that removes the ambiguiuty of natural-languages.

References

[1] Daniel Jackson. Alloy: a lightweight object modelling notation. *ACM Transactions on software engineering and methodology (TOSEM)*, 11(2):256–290, 2002.

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